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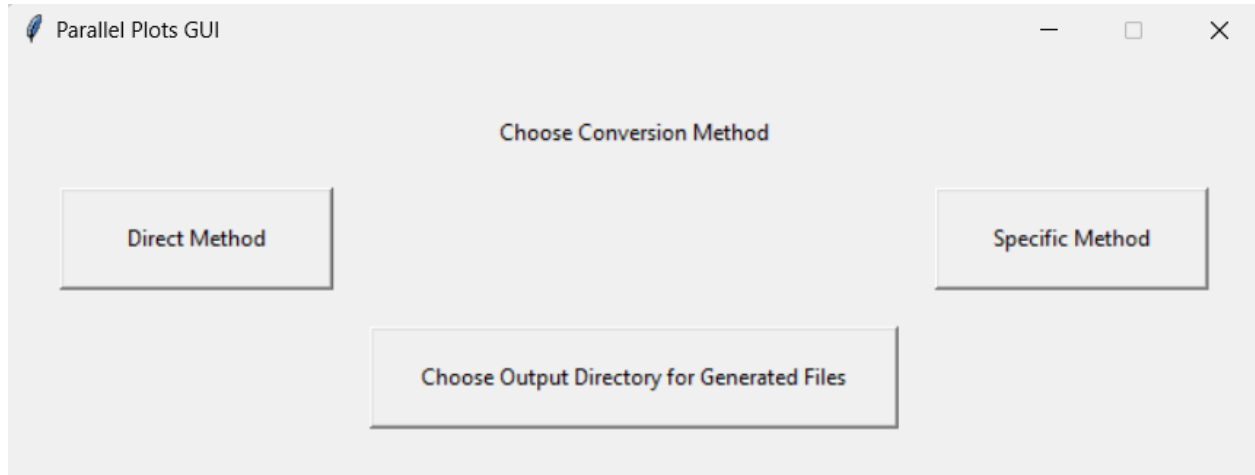
- Equations for a/b waves, Q-factor,

Installation Instructions

- Python, packages, etc

1. MAIN MENU

Upon launching the GUI, the first screen you should encounter is as follows:



From here, you can choose between the direct and specific methods for generating tabular data for parallel plots. Additionally, here one can select the output directory for all generated/exported tabular files in .txt format. If an output directory is not selected, all generated files will appear in the same directory as the Python file for the GUI.

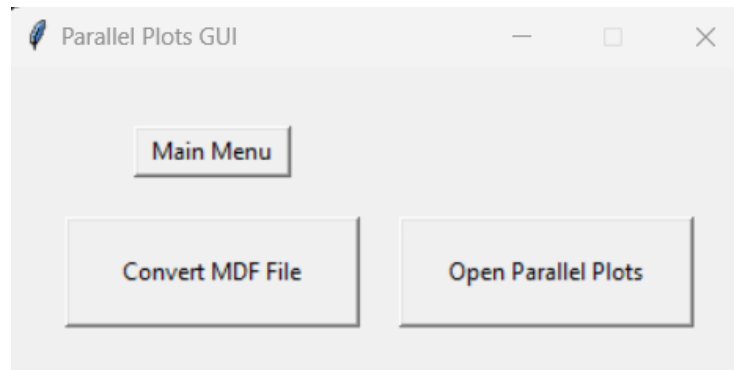
Any main menu button seen at any other point in the GUI will take you back to this screen.

2. DIRECT METHOD

The direct method is intended for generating tabular data directly from a .MDF file created via Load Pull simulation in AWR. This expedites the process of being able to go from Load Pull to Parallel Plots without having to create and export tabular files from AWR manually. The drawback is that not all measurement types are supported, as they each have to be calculated manually from A/B waves. The currently supported measurement types are:

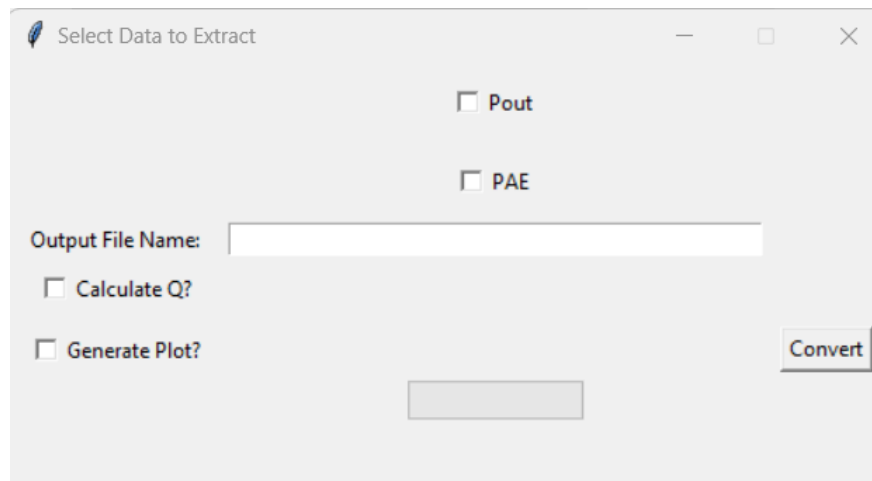
- PAE (Power Added Efficiency)
- Output Power

Choosing this method from the main menu will yield the following screen:



From here, one can launch the Parallel Plots application in a new browser window by clicking “Open Parallel Plots” or begin an MDF conversion by clicking “Convert MDF File”. This will open a file explorer prompting for the selection of an MDF to convert.

After selecting an MDF file, the following menu will appear:



From this menu, the desired measurement types can be selected and deselected at the user's discretion. One tabular file will be generated for each measurement selected, and will be in the directory specified by the user in the main menu, if applicable.

An output file name can be optionally provided to dictate the naming convention of the tabular files. If only one measurement is selected, the tabular file's name will be identical to what is specified in the entry box. If multiple measurements are selected, the tabular files will contain the name followed by the measurement type, separated by an underscore. If no file name is provided, the tabular files will be named identically to the source MDF followed by the word "Converted" and the measurement name, if relevant.

If "Calculate Q" is selected, a notation will be made in the tabular file that indicates to the Parallel Plots application to calculate Q-factor for both the source and load. Note that the tabular file will remain virtually identical and Q-factor data will NOT be present in the tabular file, as this is calculated when uploading to the Parallel Plots application.

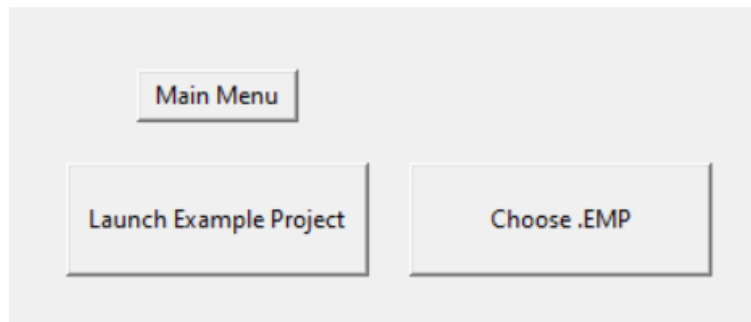
If "Generate Plot" is selected, a preliminary version of the parallel plots will be generated using Python's paxplot library for the last measurement selected. While this does not allow for the solution checking and intractability of the primary application, this can be used to verify that the correct MDF was selected and the tabular data was generated correctly as a sanity check.

For larger MDF files, the progress bar will be a useful indicator of how much of the file has been converted to tabular. Additionally, an estimate for the time remaining for the conversion will display for exceptionally large MDF files. Note that the progress bar and time remaining is an indicator for the current file being worked on, which corresponds to a single measurement type. If multiple measurement types are selected, there will be one full iteration of the progress bar per measurement and it will reset for the next one.

3. SPECIFIC METHOD

The specific method is intended for simulations in circumstances where a normal Load Pull will not be applicable in AWR. This can include simulations on schematics with multi-stage designs, sensitivity analysis, and more. Additional measurement types to PAE and Output Power can also be simulated as the parallel plot code will separate out the individual column types (PGain, Itime, Vharm, etc). Specifically, the GUI provides an easy interface for simulating sweeps for different schematics in an AWR project.

Upon choosing Specific Method, the user is presented with the following menu:



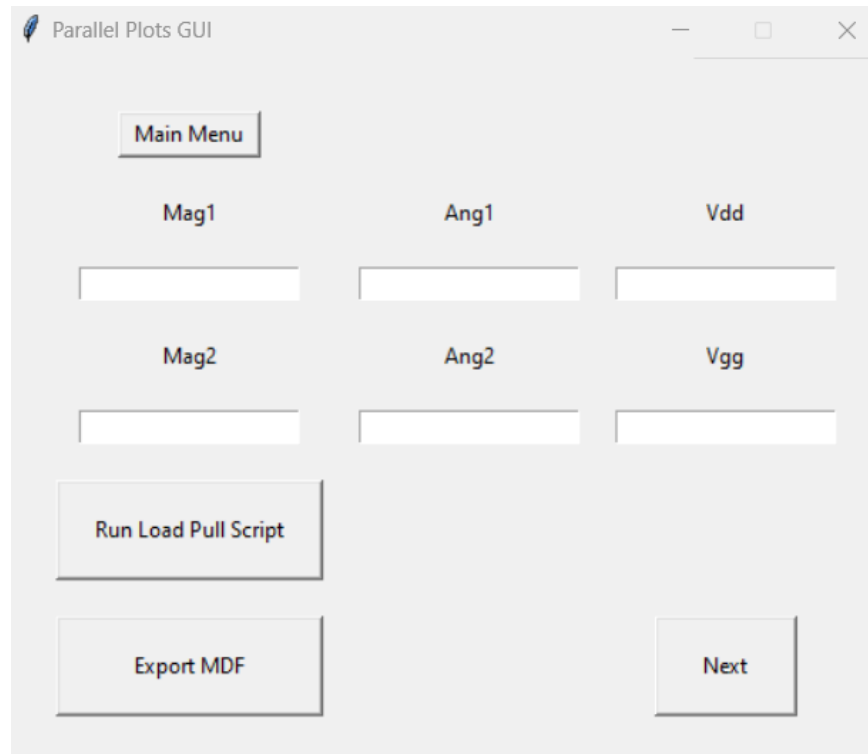
Clicking “Launch Example Project” will attempt to open “Parallel_Plot_Example_File.emp” in AWR. If this file does not exist, the operation will fail.

Alternatively, the user can click “Choose .EMP” to select an AWR project file to launch AWR with, and the GUI will bring the user directly to the sweeps interface.

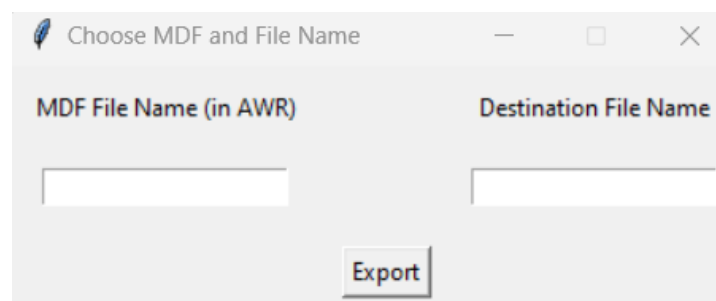
Note that this step needs to be done for all GUI interactions with AWR. There is no way for the GUI to detect an instance of AWR that is already open with a desired project.

3a. EXAMPLE PROJECT

If launching the example project, the user will be presented with the following screen:



This menu allows the user to specify values for the relevant variables in the “Load_Pull_Template” schematic, and run a Load Pull simulation by clicking “Run Load Pull Script”. Clicking this button will populate the specified values from the GUI into the schematic and then launch AWR’s Load Pull tool. Once the desired Load Pull has been simulated, the user can export the corresponding MDF, which is helpful if the user wants to use the results of this simulation in the direct method provided by the GUI:



Clicking the “Next” button will bring the user to the sweeps interface, which is where the user is originally brought if selecting their own project in the specific method.

3b. SWEEPS INTERFACE

The screenshot displays the 'Parallel Plots GUI' window. At the top left, there is a 'Manual SP and LP' button with a refresh icon. To its right is an 'Open Parallel Plots' button. Below these is a 'Repopulate Sweep Variables' button. The main area contains a table with three columns: 'Start', 'End', and 'Step'. Each row represents a variable to be swept, with a checkbox to its left. The variables listed are: 'Mag_Gamma_Source', 'Ang_Gamma_Source', 'Mag_Gamma_Load', 'Ang_Gamma_Load', 'iVd', and 'iVg'. Below the table, there is a 'Tabular Graph Name:' label followed by a text input field. To the right of this is a 'Run Manual Sweep' button. Further right is a 'Tabular File to Export:' label followed by a dropdown menu currently showing 'PAE_Tabular'. Below the 'Run Manual Sweep' button is a 'Modify Measurement' button. To the right of this is a 'File Name:' label followed by a text input field. At the bottom left is a 'Previous' button, and at the bottom right is a 'Create File' button.

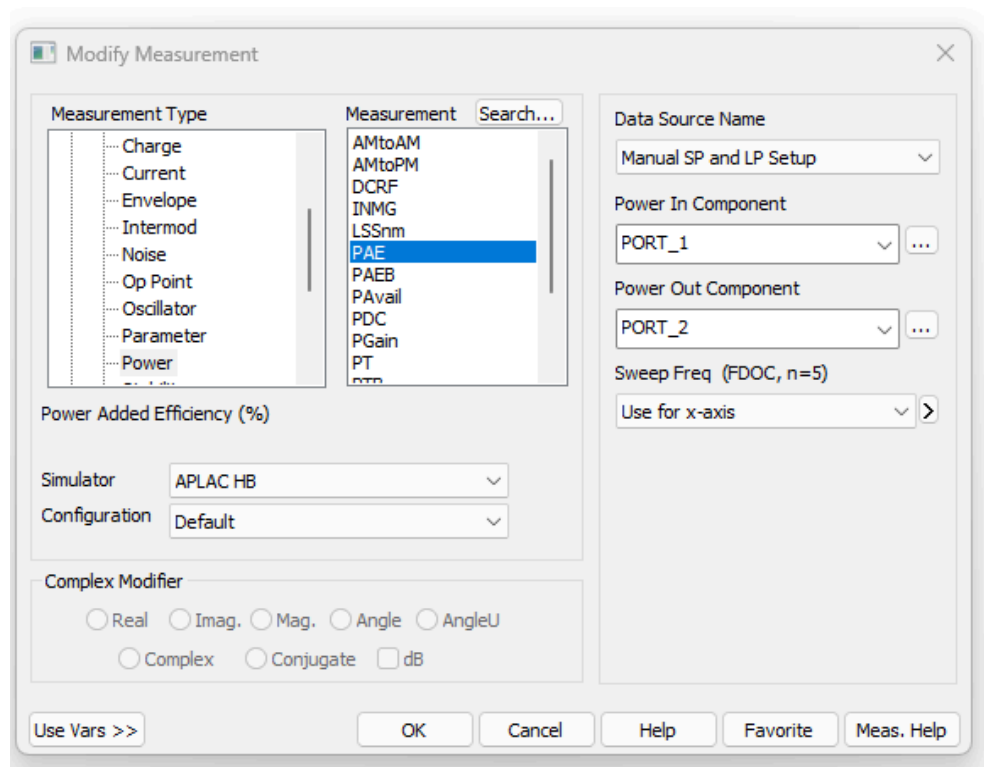
	Start	End	Step
☐ "Mag_Gamma_Source"			
☐ "Ang_Gamma_Source"			
☐ "Mag_Gamma_Load"			
☐ "Ang_Gamma_Load"			
☐ "iVd"			
☐ "iVg"			

The sweeps interface allows the user to choose between any of the schematics in their currently open AWR project and create sweeps for the relevant variables in each schematic. Note that these entries are currently determined by retrieving all the “SWPVAR” elements from each schematic. If the user wants to create another variable to sweep, the user needs to add another SWPVAR element to their AWR schematic and then click “Repopulate Sweep Variables” in the GUI to see the new corresponding row.

If the user wants to create a new schematic in their project and have it be visible in the GUI, one simply needs to click the refresh icon next to the schematic dropdown list after the schematic is created and the new schematic will be inserted into the list.

The checkboxes will allow the user to dynamically enable and disable SWPVARs in the currently selected schematic, which allows the user to quickly specify a variety of different sweeps to simulate in AWR. Once the user is satisfied with their chosen values, clicking “Run Manual Sweep” will perform the simulation in AWR. This is the direct equivalent of clicking the “Simulate” button in AWR.

A new tabular file will be generated at the completion of the simulation. If the user optionally provided a name, the tabular file will be named exactly as specified by the user. Otherwise, by default it will be named “Tabular 1” and all subsequently created tabular files will follow this same naming convention (“Tabular 2”, “Tabular 3”, etc.) By default, the measurement type will be PAE for the first created tabular file, and this can be subsequently changed by clicking on the “Modify Measurement” button. Depending on the new measurement, this may cause the program to run a new simulation.



After any new simulation performed by the GUI, the corresponding tabular file should appear in the dropdown in the bottom right of the menu. From here, it can be selected and directly exported to a .txt file with an optionally specified name. If a name is not specified, it will retain the same name as displayed in AWR.

Note that the tabular data will be for the most recent measurement simulated for that given file in AWR. The most efficient method for generating multiple tabular text files is probably to run a

simulation, export the file, modify the measurement and repeat, recycling the same tabular file in AWR for each measurement. Alternatively, a new AWR tabular file can be created for each different measurement type by clicking “Run Manual Sweep” and then modifying the measurement for each one. This is most useful if wanting to preserve the tabular data for each measurement in AWR for future use.

As in the direct method, a button is also present at the top right to allow the user to launch the Parallel Plots application in a new browser window.

4. PARALLEL PLOTS

The user should see the following upon attempting to launch the Parallel Plots application:

Multi-Objective Transistor Performance

Upload AWR Files	Column Specification	Robustness	Non-Domination	Export
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1. Upload Files

Drag and Drop Here to upload!

Convert AWR Format

The user can either drag and drop files from an existing window to upload them, or click directly on the upload box to open a file explorer, prompting for the selection of the tabular text files. Up to 5 files can be selected for inclusion in the generated parallel plots.

After selection, wait until the files appear as uploaded on the right side of the screen before continuing:

Multi-Objective Transistor Performance

Upload AWR Files	Column Specification	Robustness	Non-Domination	Export
------------------	----------------------	------------	----------------	--------

1. Upload Files

Uploaded: test_tab_pout.txt

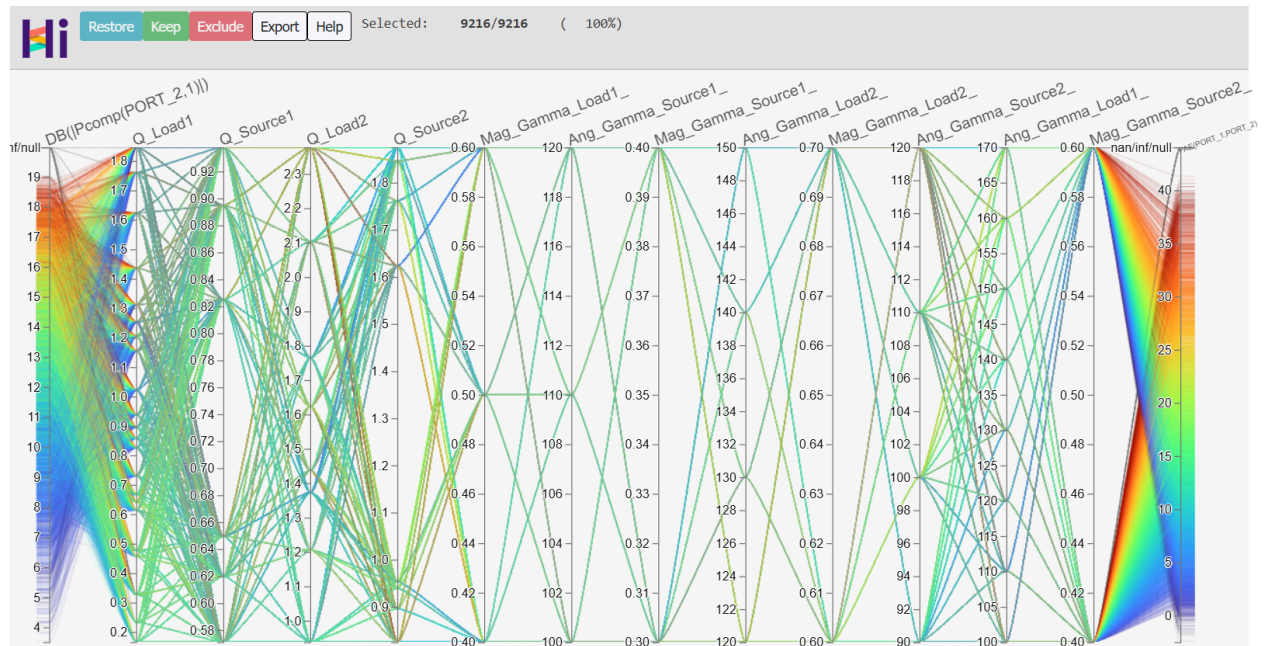
- Uploaded: uploads\b127ddd-384b-11ef-8431-c26066bdc139\test_tab_pae.txt
- Uploaded: uploads\b127ddd-384b-11ef-8431-c26066bdc139\test_tab_pout.txt

Convert AWR Format

Once the files are uploaded, click on the “Convert AWR Format” button and wait until the word “Done” appears in the button. During this process, the browser tab should say “Updating...” while it is working, although there will be no other visual indication on the screen that the application hasn’t crashed or thrown an error.

Convert AWR Format: Done

Once the conversion is done, the user can click on “Column Specification” to view the parallel plots and make selections for which columns will serve as the decisions, states, and objectives.



Once the user has selected their columns, the 3rd tab can be used to compute robustness, and the 4th tab can be used to compute all non-dominated solutions. Lastly, the user can export the computed data so it can be directly opened as an .HTML file later without having to upload the tabular data and make the column selections again.

MDIF Conversion Slides

.mdf file contains, a and b waves, V_{gg}, V_{dd}, igg, idd.

$$mag = \sqrt{Re^2 + Im^2}$$

$$Power_{dBm} = 10 * 10 \log(Power_{Watts}) + 30$$

$$P_{in} = \frac{mag(a_1)^2}{2}$$

$$Power_{watts} = \frac{10^{\frac{power_{dBm}}{10}}}{1000}$$

$$P_{del} = \frac{mag(a_1)^2 - mag(b_1)^2}{2}$$

$$Q = \frac{Imag(Z)}{Real(Z)}$$

$$Gamma = \frac{mag(a_i)}{mag(b_i)}$$

$$P_{out} = \frac{mag(b_2)^2 - mag(a_2)^2}{2}$$

$$Gamma_{Load} = \frac{mag(a_2)}{mag(b_2)}$$

$$P_{dc} = V_{dd} * I_{dd}$$

$$Gamma_{Source} = \frac{mag(a_1)}{mag(b_1)}$$

$$PAE = \frac{(P_{out}(Watts) - P_{in}(Watts)) * 100}{P_{dc}}$$

$$ang(a_i) = \tan^{-1} \frac{im(a_i)}{re(a_i)}$$

$$PAE(del) = \frac{(P_{out}(Watts) - P_{del}(Watts)) * 100}{P_{dc}}$$

$$angle = ang(a_i) - ang(b_i) + 360$$

Installation Instructions:

This application was tested on Python 3.9. If you experience any issues with packages on later versions of Python, consider downgrading to Python 3.9.

If you don't have python installed, download Version 3.9 with the link below.

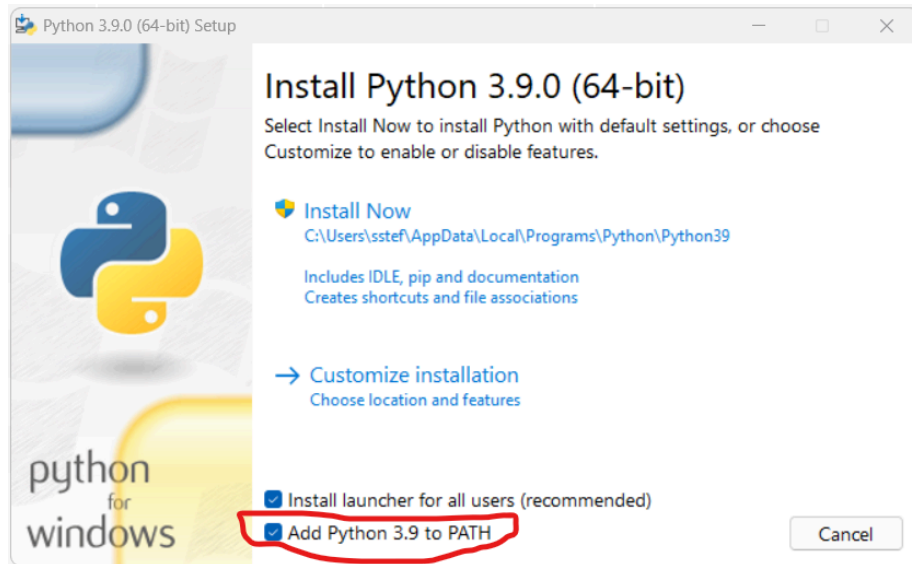
<https://www.python.org/downloads/release/python-390/>

1. Download the file type that matches your computer

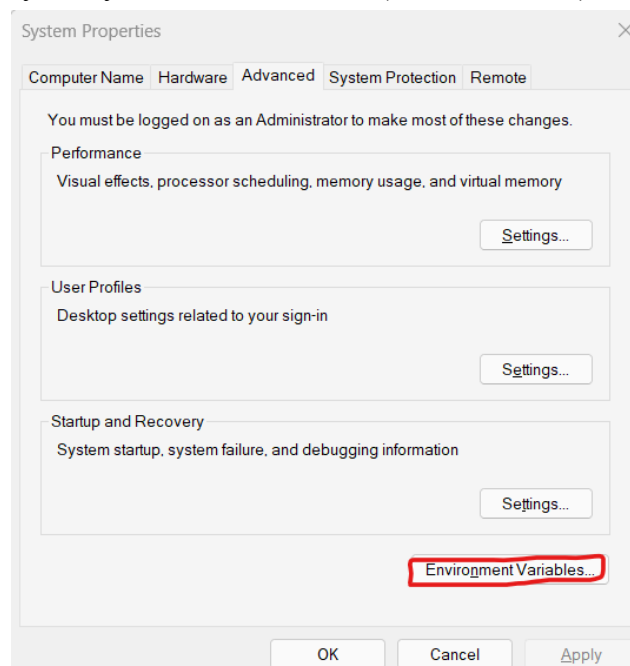
Files					
Version	Operating System	Description	MD5 Sum	File Size	GPG
Gzipped source tarball	Source release		e19e75ec81dd04de27797bf3f9d918fd	25.5 MB	SIG
XZ compressed source tarball	Source release		6ebfe157f6e88d9eabfbaf3fa92129f6	18.0 MB	SIG
macOS 64-bit installer	macOS	for OS X 10.9 and later	16ca86fa3467e75bade26b8a9703c27f	29.7 MB	SIG
Windows help file	Windows		9ea6fc676f0fa3b95af3c5b3400120d6	8.4 MB	SIG
Windows x86 embeddable zip file	Windows		d81fc534080e10bb4172ad7ae3da5247	7.2 MB	SIG
Windows x86 executable installer	Windows		4a2812db8ab9f2e522c96c7728cfcccb	25.8 MB	SIG
Windows x86 web-based installer	Windows		cdbfa799e6760c13d06d0c2374110aa3	1.3 MB	SIG
Windows x86-64 embeddable zip file	Windows	for AMD64/EM64T/x64	60d0d94337ef657c2cca1d3d9a6dd94b	8.0 MB	SIG
Windows x86-64 executable installer	Windows	for AMD64/EM64T/x64	b61a33dc28f13b561452f3089c87eb63	26.9 MB	SIG
Windows x86-64 web-based installer	Windows	for AMD64/EM64T/x64	733df85afb160482c5636ca09b89c4c8	1.3 MB	SIG

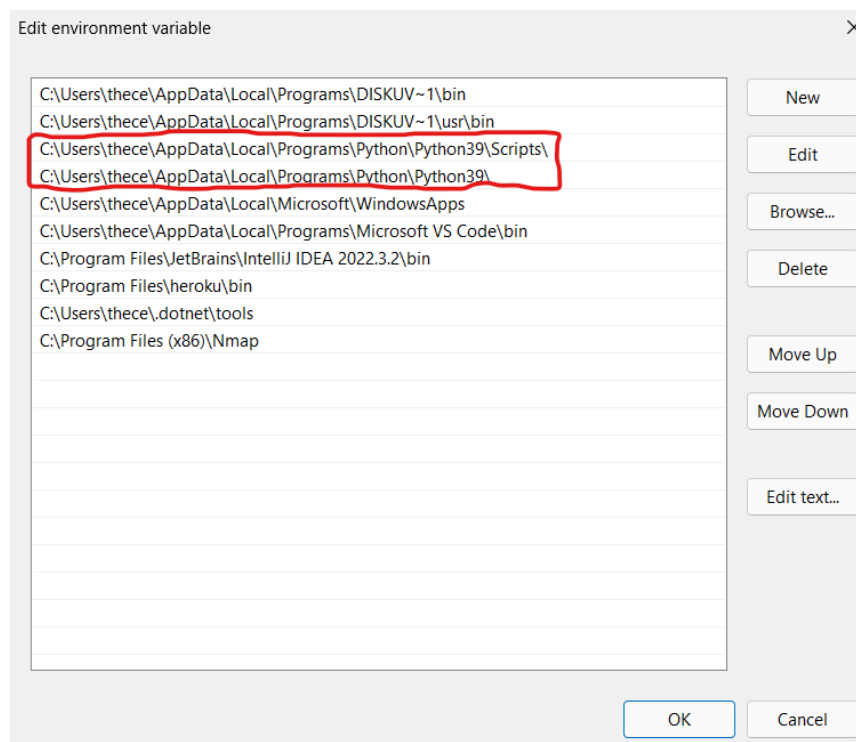
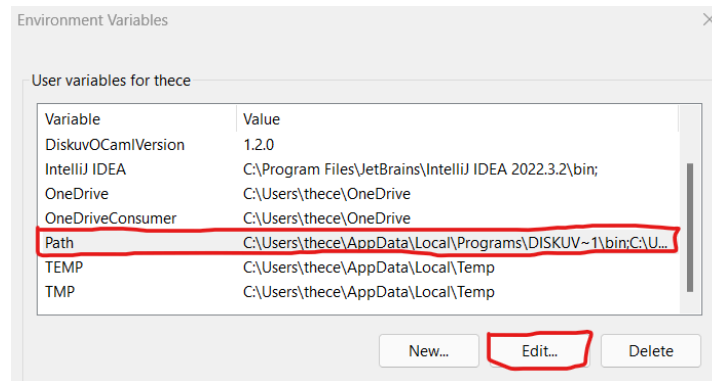
2. Should most likely be Windows x86-64 executable installer

When downloading python 3.9, make sure to click “Add Python 3.9 to PATH”



If you have multiple versions of Python installed, make sure your PATH environment variable points to the version of Python you want to run with (in this case 3.9)

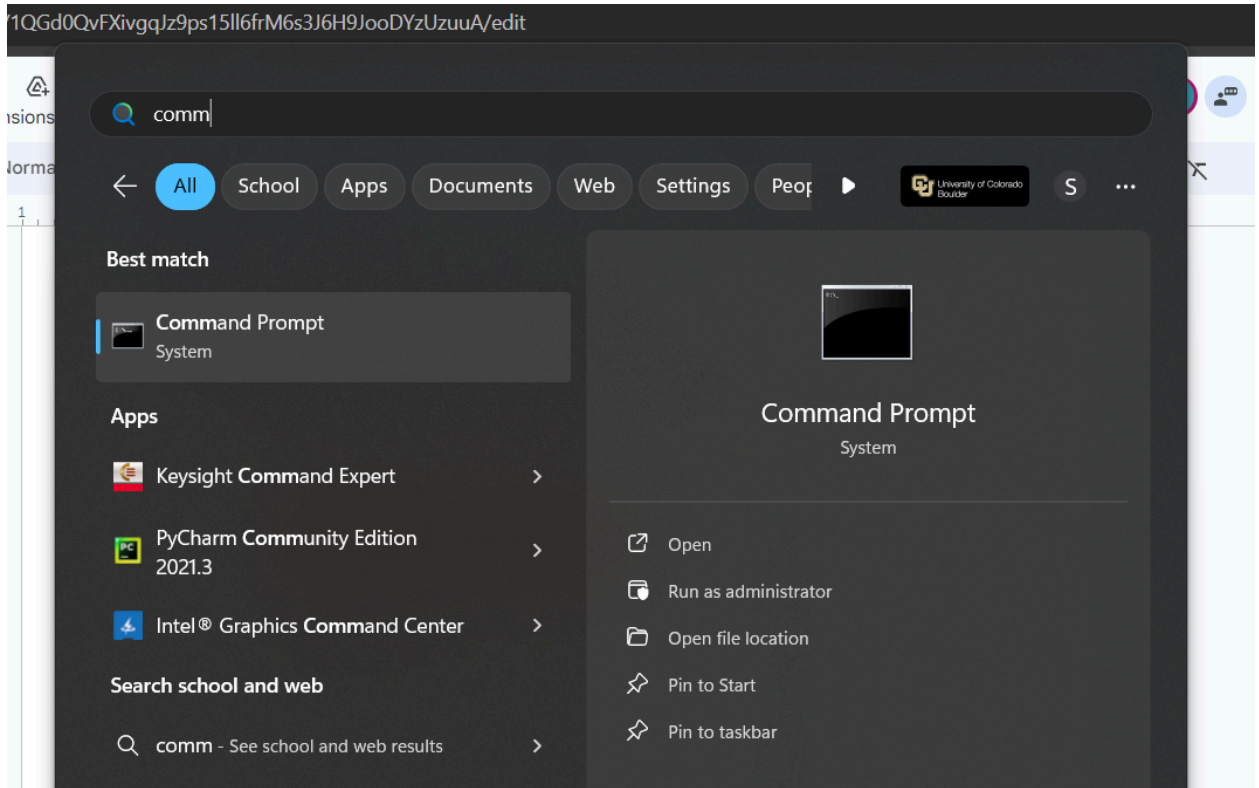




If the desired version of Python is not contained within the PATH environment variable, you can point to the correct directory by clicking “Add” and creating a new entry. If you want to change versions of Python, you simply need to “Edit” the corresponding entries and change the value to the correct directory. In my case, it would be changing “Python39” to “Python311” or something equivalent.

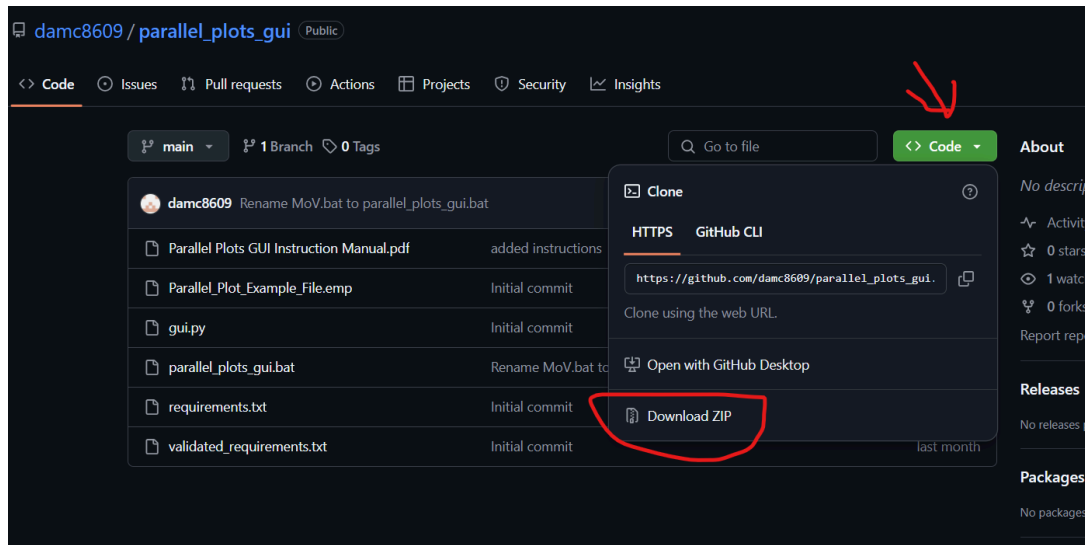
The following instructions assume you already have a valid Python installation, including pip, as well as a git installation.

1. Open a command prompt terminal

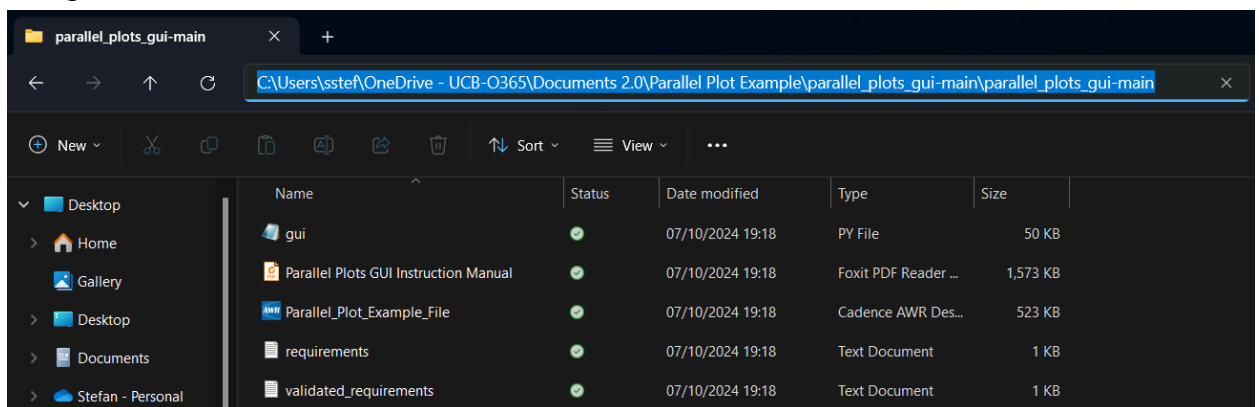


2. Clone into the following Git repository: https://github.com/damc8609/parallel_plots_gui. This can be done as follows **or** you can download the zip file and extract to a directory.

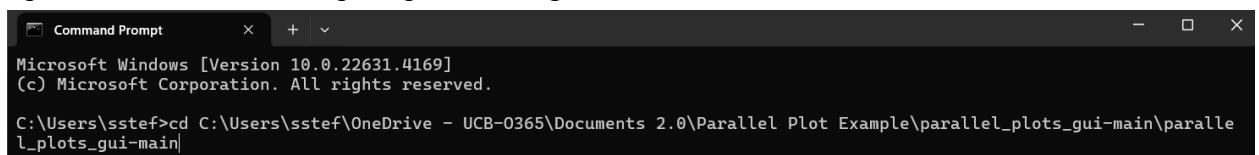
```
C:\Users\thece\Downloads\rf_research_summer_2023\gui_project\install>git clone https://github.com/damc8609/parallel_plots_gui
Cloning into 'parallel_plots_gui'...
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (7/7), done.
remote: Total 9 (delta 1), reused 9 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (9/9), 2.02 MiB | 6.10 MiB/s, done.
Resolving deltas: 100% (1/1), done.
C:\Users\thece\Downloads\rf_research_summer_2023\gui_project\install>
```

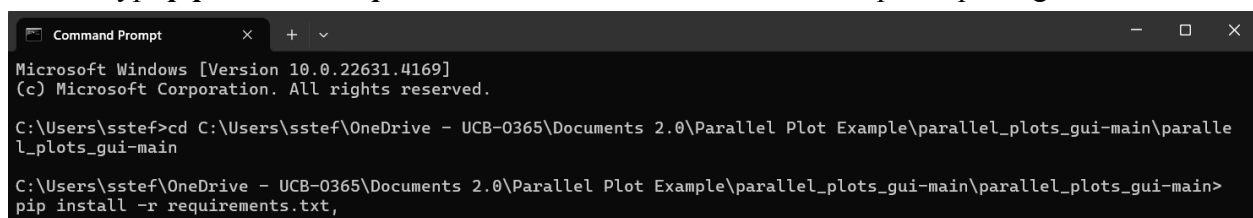
3. Navigate to the installed folder



4. Open windows command prompt and navigate to the correct folder



5. Type `pip install -r requirements.txt` which will install all required packages.



If you run into this, you either do not have python installed, or you do not have python on the system path.

```
'pip' is not recognized as an internal or external command,
operable program or batch file.
```

This should start downloading then as shown below

```
C:\Users\sstef>cd C:\Users\sstef\OneDrive - UCB-0365\Documents 2.0\Parallel Plot Example\parallel_plots_gui-main\parallel_plots_gui-main

C:\Users\sstef\OneDrive - UCB-0365\Documents 2.0\Parallel Plot Example\parallel_plots_gui-main\parallel_plots_gui-main>
pip install -r requirements.txt
Collecting pyawr
  Downloading pyawr-24.1.1-py3-none-any.whl (156 kB)
    |#####| 156 kB 6.8 MB/s
Collecting pywin32
  Downloading pywin32-307-cp39-cp39-win_amd64.whl (6.6 MB)
    |#####| 6.6 MB 6.4 MB/s
Collecting mopa
  Using cached mopa-0.2.5b0-py3-none-any.whl (9.1 kB)
Collecting paxplot
  Downloading paxplot-0.1.4b2-py3-none-any.whl (11 kB)
Collecting packaging==21.3
  Downloading packaging-21.3-py3-none-any.whl (40 kB)
    |#####| 40 kB 2.7 MB/s
Processing c:\users\sstef\appdata\local\pip\cache\wheels\04\c5\42\8a8f19209f353d66c38380a9f0cfd4eb605eece06c4a773772\dash-uploader-0.7.0a1-py3-none-any.whl
Collecting dash-bootstrap-components
  Downloading dash_bootstrap_components-1.6.0-py3-none-any.whl (222 kB)
    |#####| 222 kB 6.4 MB/s
Collecting dash
  Downloading dash-2.18.1-py3-none-any.whl (7.5 MB)
    |#####| 7.5 MB ...
```

Once download is complete, it will look like this

```
    |#####| 103 kB 6.4 MB/s
Installing collected packages: pywin32, pyawr, zipp, importlib-metadata, urllib3, charset-normalizer, certifi, idna, requests, dash-table, nest-asyncio, tenacity, pyparsing, packaging, plotly, dash-core-components, MarkupSafe, Werkzeug, six, retrying, Jinja2, itsdangerous, blinker, colorama, click, Flask, dash-html-components, typing-extensions, dash, dash-uploader, dash-bootstrap-components, numpy, tzdata, pytz, python-dateutil, pandas, soupsieve, beautifulsoup4, exceptiongroup, traitlets, pure-eval, executing, asttokens, stack-data, matplotlib-inline, wcwidth, prompt-toolkit, parso, jedi, pygments, decorator, ipython, zstandard, brotli, flask-compress, hiplot, mopa, cycler, importlib-resources, fonttools, contourpy, pillow, kiwisolver, matplotlib, paxplot
Successfully installed Flask-3.0.3 Jinja2-3.1.4 MarkupSafe-3.0.0 Werkzeug-3.0.4 asttokens-2.4.1 beautifulsoup4-4.12.3 blinker-1.8.2 brotli-1.1.0 certifi-2024.8.30 charset-normalizer-3.3.2 click-8.1.7 colorama-0.4.6 contourpy-1.3.0 cycler-0.12.1 dash-2.18.1 dash-bootstrap-components-1.6.0 dash-core-components-2.0.0 dash-html-components-2.0.0 dash-table-5.0.0 dash-uploader-0.7.0a1 decorator-5.1.1 exceptiongroup-1.2.2 executing-2.1.0 flask-compress-1.15 fonttools-4.54.1 hiplot-0.1.33 idna-3.10 importlib-metadata-8.5.0 importlib-resources-6.4.5 ipython-8.18.1 itsdangerous-2.2.0 jedi-0.19.1 kiwisolver-1.4.7 matplotlib-3.9.2 matplotlib-inline-0.1.7 mopa-0.2.5b0 nest-asyncio-1.6.0 numpy-2.0.2 packaging-21.3 pandas-2.2.3 parso-0.8.4 paxplot-0.1.4b2 pillow-10.4.0 plotly-5.24.1 prompt-toolkit-3.0.48 pure-eval-0.2.3 pyawr-24.1.1 pygments-2.18.0 pyparsing-3.1.4 python-dateutil-2.9.0.post0 pytz-2024.2 pywin32-307 requests-2.32.3 retrying-1.3.4 six-1.16.0 soupsieve-2.6 stack-data-0.6.3 tenacity-9.0.0 traitlets-5.14.3 typing-extensions-4.12.2 tzdata-2024.2 urllib3-2.2.3 wcwidth-0.2.13 zipp-3.20.2 zstandard-0.23.0
WARNING: You are using pip version 20.2.3; however, version 24.2 is available.
You should consider upgrading via the 'c:\users\sstef\appdata\local\programs\python\python39\python.exe -m pip install --upgrade pip' command.

C:\Users\sstef\OneDrive - UCB-0365\Documents 2.0\Parallel Plot Example\parallel_plots_gui-main\parallel_plots_gui-main>
```

- a. In the event that future package releases cause compatibility issues, *validated_requirements.txt* attempts to install known working versions of each package that were used for testing the application. Run the *pip install* command on this file if there are any issues with importing the packages.

6. Run ***python gui.py*** in the terminal window to run the Python application.

```
C:\Users\sstef\OneDrive - UCB-0365\Documents 2.0\Parallel Plot Example\parallel_plots_gui-main\parallel_plots_gui-main>python gui.py
```

Alternatively, if you are on a Windows machine, the install directory contains a batch file (parallel_plots_gui.bat), that, when run as an application, should launch the GUI. This is particularly useful if you want to create a shortcut to the batch file by right-clicking on it and clicking Create Shortcut. Then, this shortcut can be added to the Desktop or anywhere else it is convenient and then the GUI can be launched at any time without having to use the command line.



Once completed and open Parallel Plots GUI

```
C:\Users\sstef\OneDrive - UCB-0365\Documents 2.0\Parallel Plot Example\parallel_plots_gui-main\parallel_plots_gui-main>python gui.py
Dash is running on http://127.0.0.1:8050/
```

```
Launching GUI
* Serving Flask app 'mopa.app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:8050
Press CTRL+C to quit
```

